

Fundamentals of Database Systems

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Chapter 1: Introduction to Database Systems

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Outline

- ✓ Introduction
- ✓ Data, Information, and Metadata
- ✓ File-Based Approach
- ✓ Database Approach
- ✓ Database Development Process
- ✓ Database Users (Actors on the scene)
- ✓ Summary

Introduction

- ✓ What do you think a database system is?
- ✓ How many of you have seen a database?
- ✓ What kind of database do you know?
- ✓ Think of some (real life) examples of database system?
- ❖ **Why do you need to study database? -->**
 - ✓ Because, everybody needs it.
 - ✓ It is an integral part of our day to day life.

Basic Definitions

- ✓ A **database** is nothing more than a collection of shared information/data that exists over a long period of time, often many years.
- ✓ **Database** is a collection of interrelated data.
What is data? Data is known facts that can be recorded and have an implicit meaning.
- ✓ The term **database** refers to a collection of data that is managed by a **DBMS**.
- ✓ **Database systems** are designed to manage large data set in an organization.

Cont..

- ✓ The database is an **integrated collection** of facts about an organization.
- ✓ Organization can be a **University** or a **department** in a University , **Insurance companies, manufacturing companies, Banks, Airlines, Telecommunications, Governmental and Non-governmental Organizations, Research institutions ,etc.**
- ✓ The **database** is used as a central data source for other applications

Data

✓ The term **data** refers to known **raw facts** about things like **people, places, events** and **concepts**:

✓ **Sources of data**: careful observation, experiment and research .

✓ **Data** can exist in the form of **text, graphics, images, audio** and/or **video**

ID	Name	Address	City	State
1	Andrew	SW 16 th Ave	Orlando	FL
2	Robert	23 rd Terrace	Boston	MA
3	Bob	52nd Avenue	Bronx	NY
4	Lee Hung	52nd Avenue	Albany	NY
5	Richard	999 Palm Bay	Detroit	MI
6	Tony	15th Avenue	Austin	TX

Data in context

Flow of Data

✓ **Data** → **Information** → **Knowledge** → **Action**

- ✓ Summarizing the Data
- ✓ Averaging the data
- ✓ Graphing the data
- ✓ Adding Context

✓ **Information** is the processed data presented in a form suitable for human interpretation.

- ✓ **The same** data can be processed in a variety of ways to generate different kinds of information.
- ✓ **Information** is important for ‘**informed decision making**’.

Metadata

➤ **Metadata** is data about data.

- ✓ For further understanding of large collection of data an additional description of the data is very important
- ✓ It increases the usability of the data – Self-describing nature of databases

Name	Data Type	Size	Description
ID	Number	4	Unique Student ID
Name	Text	50	Student name
Address	Text	200	Mailing Address
City	Text	50	Residential City
State	Text	2	Residential State
ZIP	Number	5	ZIP Code

Metadata

Database systems :- are designed to manage large data set in an organization.

Today, Databases are essential to every business.

They are used:

- ✓ To maintain internal records,
- ✓ To present data to customers and clients on the **World-Wide-Web**, and
- ✓ To support many other commercial processes.

Evolution of Database Systems

- Data management passes through the different levels of development along with the development in technology and services.
- Also the approaches to change data to information takes this level of process:we can classify into the following three basic approaches:
 - ✓ Manual approach
 - ✓ Traditional file based approach
 - ✓ Database approach

Evolution of Database Systems

➤ The Manual Approach

- ✓ In the manual approach, data storage and retrieval follows the primitive and traditional way of information handling where *cards* and *paper* are used for the purpose.

- ✓ The data storage and retrieval will be *performed using human labor*.

- Files for as many event and objects as the organization are used to store information.

- Each of the files containing various kinds of information is labeled and stored in one or more **cabinets**.

The Manual Approach (Continued..)

➤ Limitations of the Manual Approach

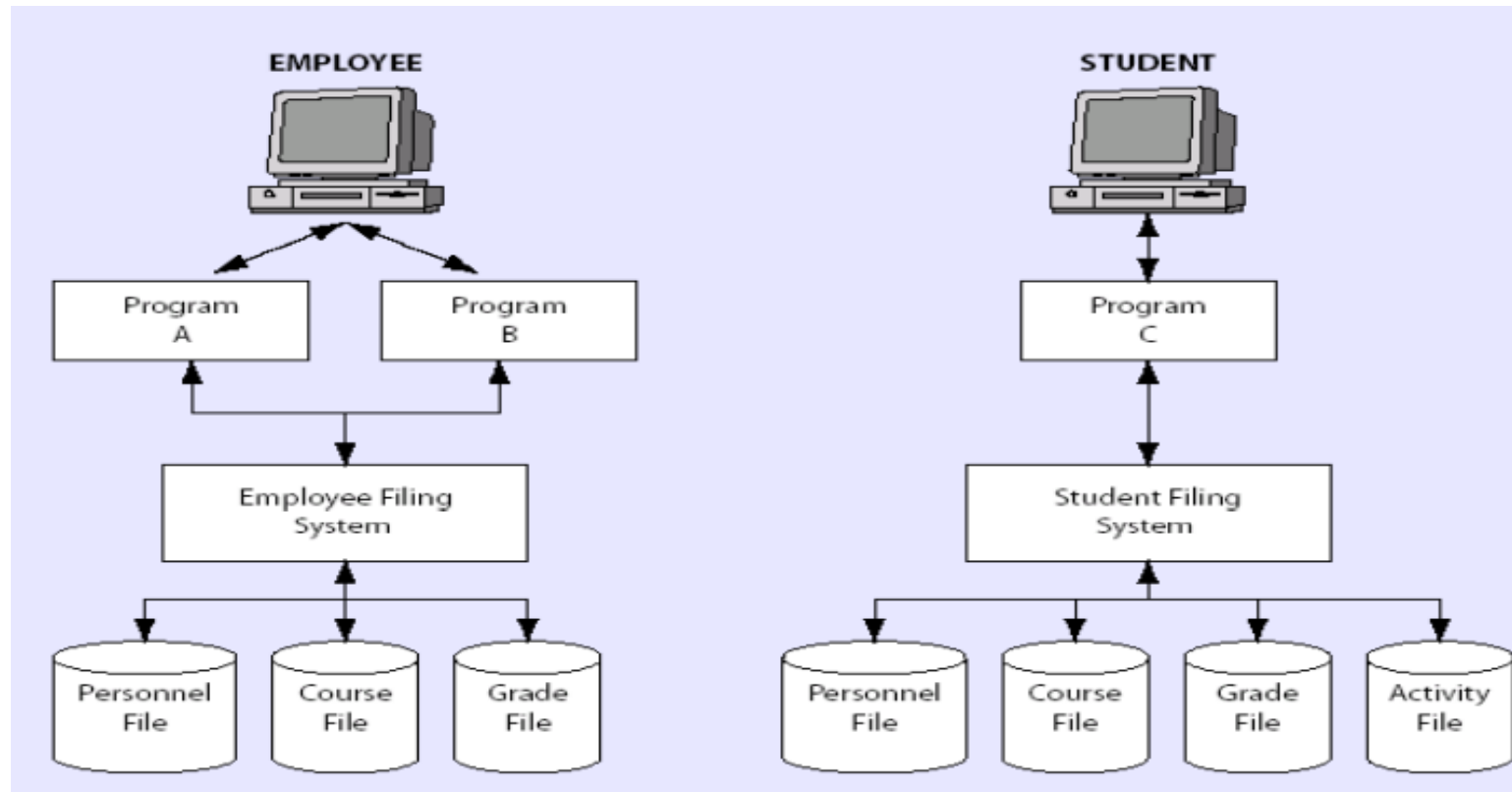
- ✓ Prone to error
- ✓ Difficult to update, retrieve, integrate
- ✓ You have the data but it is difficult to compile the information
- ✓ Limited to small size information
- ✓ Cross referencing is difficult

Traditional File-Based Approach

- ✓ File based systems were an early attempt to computerize the manual filing system. This approach is the decentralized computerized data handling method.
- ✓ Since every application defines and manages its own data, the system is subjected to serious data duplication problem

cont...

❖ Example:



University
File-Based System

Limitations of File-Based Approach

- ❖ Much efforts for adhoc query answering:
 - ✓ What is the average grade for Mr. Said's students?
 - ✓ List the activities for all students enrolled in CSE309.
 - ✓ Which personnel are students as well as staff?
- ❖ Other limitations:
 - ✓ Duplication of data
 - ✓ Data dependency
 - ✓ Slow development, high maintenance and fixed queries.

Cont..

The most significant problem experienced by the **traditional file based** approach of data handling are the following **three** types of **anomalies**;

- ✓ **Modification Anomalies**: a problem experienced when one or more data value is modified on one application program but not on others containing the same data set.
- ✓ **Deletion Anomalies**: a problem encountered where one record set is deleted from one application but remain untouched in other application programs.
- ✓ **Insertion Anomalies**: a problem experienced whenever there is new data item to be recorded, and the recording is not made in all the applications.

Database Approach

- **Database** is just a computerized record keeping system or a kind of electronic filing cabinet.
- **It** is a shared collection of logically related data designed to meet the information needs of an organization. Since it is a shared corporate resource, the database is integrated with minimum amount of or ***no duplication***.
- **Database** is a collection of logically related data where these logically related data comprises **entities, attributes, relationships, and business rules** of an organization's information.

Cont..

- Limitations of **file-based** approach tell us that: —
 - ✓ Parameters defining data should be separately stored.
 - ✓ There should be a way to control and manipulate data in isolation of the application program.
- A **database system** can be defined as the combination of a **database** and a **DBMS**.
- **DBMS** is a powerful *tool or Software for creating and manipulating database*.
- *Examples of DBMS:*
 - ✓ Oracle
 - ✓ MS SQL Server
 - ✓ MS Access
 - ✓ PostgreSQL
 - ✓ MySQL

Typical DBMS Functionalities

- **Define** a particular database in terms of its **data types**, **structures**, and **constraints**.
- **Construct** or **Load** the initial database contents on a secondary storage medium.
- **Manipulating** the database:
 - **Retrieval**: Querying, generating reports
 - **Modification**: Insertions, deletions and updates to its content
 - **Accessing** the database through Web applications
- **Processing** and **Sharing** by a set of concurrent users and application programs – yet, keeping all data valid and consistent.

- **Other features:**

- ✓ Protection or Security measures to prevent unauthorized access
- ✓ Presentation and Visualization of data
- ✓ Maintaining the database and associated programs over the lifetime of the database application: Called database, software, and system maintenance.

Because of the advantages offered by a DBMS, businesses and organizations prefer the database approach than the file-based approach.

Why Database Approach?

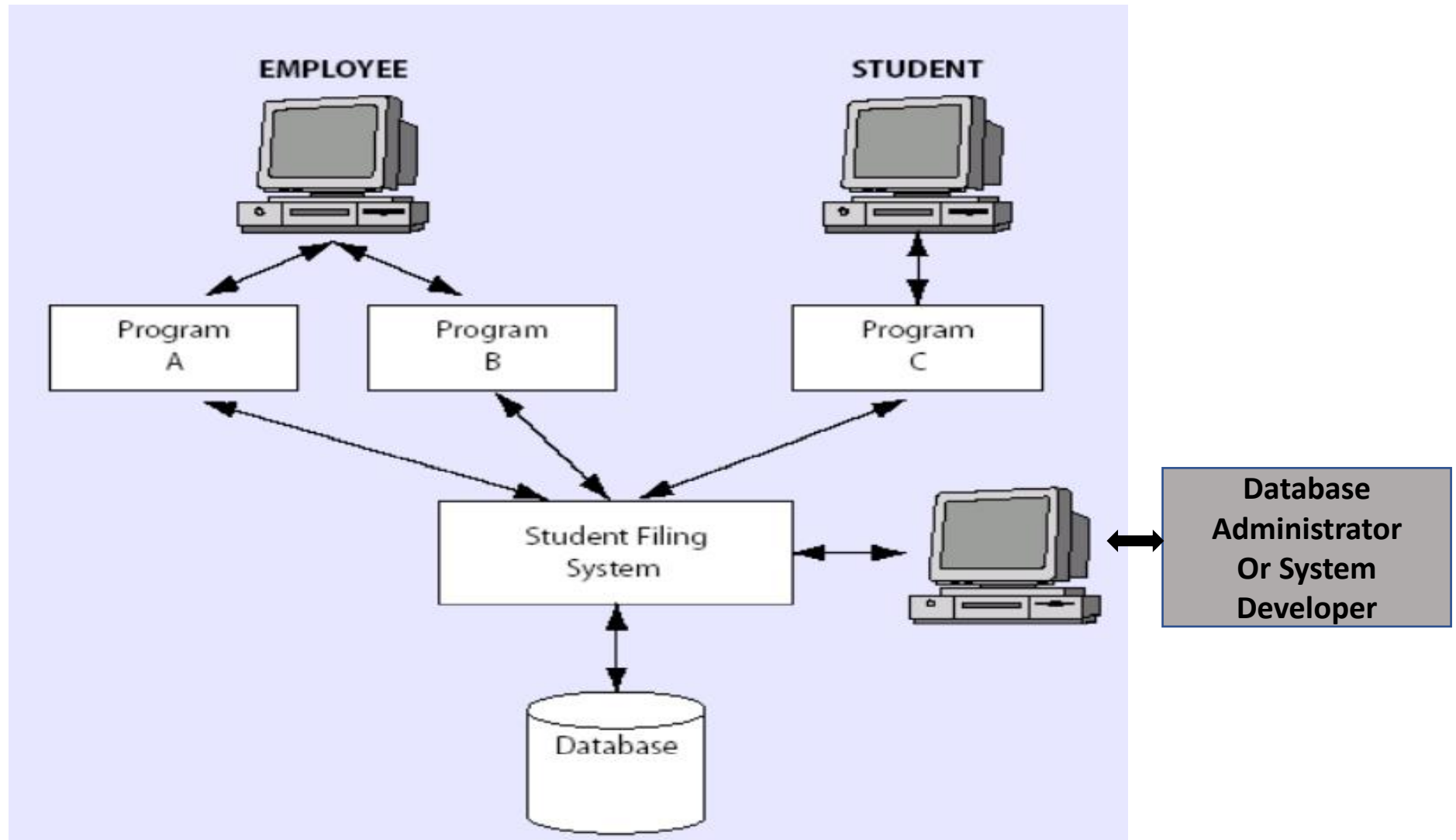
- ✓ User requirements are constantly changing
- ✓ Ad hoc questions need rapid answers
- ✓ Need to reduce long lead times and high cost in new application development. Lead time is the latency between the initiation and execution of a process.
- ✓ Data sharing throughout the organization
- ✓ Need to improve data consistency and access control
- ✓ Ease of application development and reduced development time

Cont..

- A database system consists of:
 - ✓ Data (the database)
 - ✓ Software
 - ✓ Hardware
 - ✓ Users
- We focus mainly on the software
- Database systems allow users to:
 - ✓ *Store*
 - ✓ *Update*
 - ✓ *Retrieve*
 - ✓ *Organize*
 - ✓ *Protect*

their data.

Example: University's Database System



University Database (Cont..)

- Mini-world for a University Database example: Part of a UNIVERSITY environment.

Some mini-world entities:

- *Students*
- *PRE-REQUISITE Courses*
- *Courses*
- *Departments*
- *Instructors*

University Database (Cont..)

❖ *Some mini-world relationships:*

- STUDENTs take COURSEs
- COURSEs have PRE-REQUISITE COURSEs
- INSTRUCTORs teach COURSEs
- COURSEs are offered by DEPARTMENTs
- STUDENTs major in DEPARTMENTs

*Note: The above (Entities and Relationship) could be expressed in the **Entityrelationship** data model.*

Simple Database Example

A database that stores student and course information.

COURSE

Course_name	Course_number	Credit_hours	Department
Intro to Computer Science	CS1310	4	CS
Data Structures	CS3320	4	CS
Discrete Mathematics	MATH2410	3	MATH
Database	CS3380	3	CS

SECTION

Section_identifier	Course_number	Semester	Year	Instructor
85	MATH2410	Fall	04	King
92	CS1310	Fall	04	Anderson
102	CS3320	Spring	05	Knuth
112	MATH2410	Fall	05	Chang
119	CS1310	Fall	05	Anderson
135	CS3380	Fall	05	Stone

GRADE_REPORT

Student_number	Section_identifier	Grade
17	112	B
17	119	C
8	85	A
8	92	A
8	102	B
8	135	A

PREREQUISITE

Course_number	Prerequisite_number
CS3380	CS3320
CS3380	MATH2410
CS3320	CS1310

Main Characteristics of the Database Approach

- ***Self-describing nature of a database system:*** A DBMS catalog stores the description of the database. (The description is called meta-data). This allows the DBMS software to work with different databases.
- ***Insulation between programs and data:*** Called program-data independence. Allows changing data storage structures and operations without having to change the DBMS access programs.
- ***Data Abstraction:*** A data model is used to hide storage details and present the users with a conceptual view of the database.

- ❖ **Support of multiple views of the data:** Each user may see a different view of the database, which describes only the data of interest to that user.
- ❖ **Sharing of data and multi-user transaction processing:** allowing a set of concurrent users to retrieve and to update the database.
- ❖ **Concurrency control** within the DBMS guarantees that each transaction is correctly executed or completely aborted.
- ❖ **OLTP** (Online Transaction Processing) is a major part of database applications.

Benefits of Using Database Approach

- ✓ ***Data can be shared***: two or more users can access and use same data instead of storing data in redundant manner for each user.
- ✓ ***Improved accessibility of data***: by using SQL, the users can easily access data without programming experience.
- ✓ ***Redundancy can be reduced***: isolated data is integrated in database to decrease the redundant data stored at different applications.
- ✓ ***Quality data can be maintained***: the different integrity constraints in the database approach will maintain the quality leading to better decision making.

Cont..

- ✓ ***Transaction support can be provided***: basic demands of any transaction support systems are implanted in a full scale DBMS.
- ✓ ***Integrity can be maintained***: data at different applications will be integrated together with additional constraints to facilitate shared data resource.
- ✓ ***Security majors can be enforced***: the shared data can be secured by having different levels of clearance and other data security mechanisms.
- ✓ ***Improved decision support***: the database will provide information useful for decision making.

Cont..

- ✓ ***Standards can be enforced***: the different ways of using and dealing with data by different units of an organization can be balanced and standardized by using database approach.
- ✓ ***Compactness***: since it is an electronic data handling method, the data is stored compactly (no voluminous papers).
- ✓ ***Speed***: data storage and retrieval is fast as it will be using the modern fast computer system.
- ✓ ***Less labor***: unlike the other data handling methods, data maintenance will not demand much resource.
- ✓ ***Centralized information control***: since relevant data in the organization will be stored at one repository, it can be controlled and managed at the central level.

Disadvantages of Using the Database Approach

- ✓ Introduction of new professional and specialized personnel.
- ✓ Complexity in designing and managing data
- ✓ The cost and risk during conversion from the old to the new system
- ✓ High cost to be incurred to develop and maintain the system
- ✓ Complex backup and recovery services from the users perspective
- ✓ High impact on the system when failure occurs to the central system.

When Not to Use a DBMS?

- ***Main inhibitors (costs) of using a DBMS:***
 - ✓ High initial investment and possible need for additional hardware.
 - ✓ Overhead for providing generality, security, concurrency control, recovery, and integrity functions.
- ***When a DBMS may be unnecessary:***
 - ✓ If the database and applications are simple, well defined, and not expected to change.
 - ✓ If there are stringent real-time requirements that may not be met because of DBMS overhead.
 - ✓ If access to data by multiple users is not required.

Cont..

- *When no DBMS may suit/suffice :*
 - If the database system is not able to handle the complexity of data because of modeling limitations
 - If the database users need special operations not supported by the DBMS.

Database Development Process

1. Enterprise modeling

- Describes the data needed for database system
- The real world is so complex and modeling an enterprise , or even a part of it is not an easy task
- Background knowledge of the organization is very important

2. Conceptual database modeling:

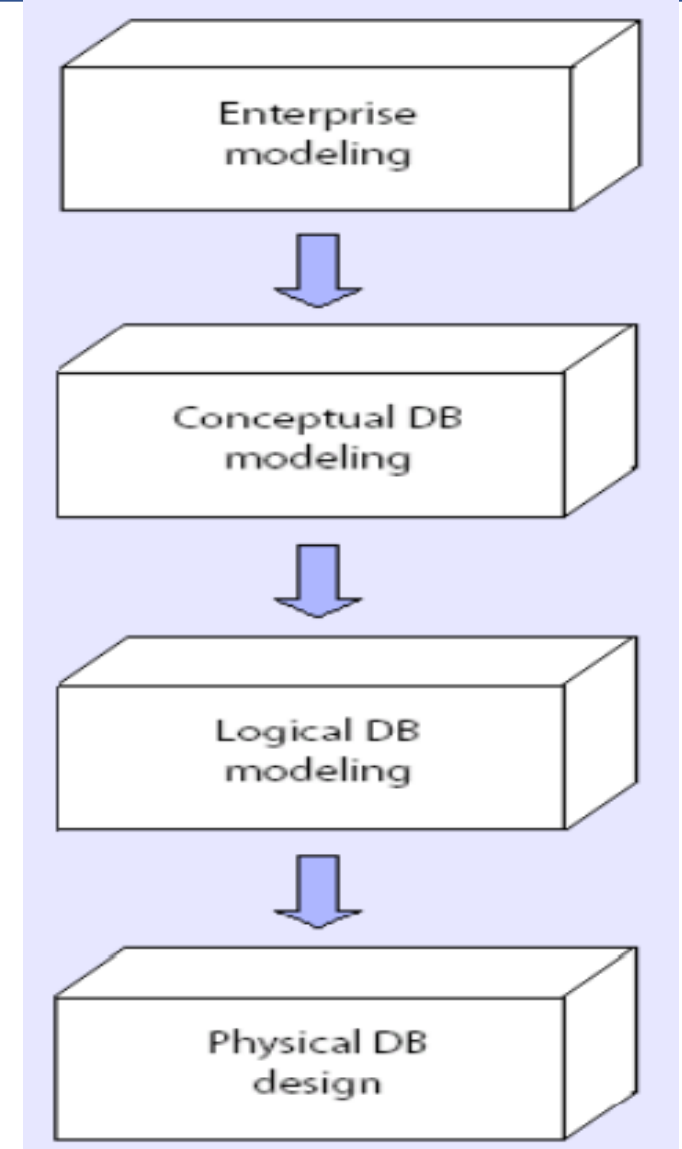
- Entity-relationship (E-R) modeling

3. Logical database design:

- Relational data modeling is used to transform the E-R diagram into a relational schema

4. Physical database design and creation:

- Detailed specification of data elements, data types, indexing options ,etc.
- Selection of software (DBMS) and hardware



Users Involved in Database Systems

❖ *They can be divided into two:*

- Actors on the Scene
- Workers behind the Scene

Actors on the Scene

- Those who actually use and control the content of the data in the database
- They include the DBA, DB Designers, End Users, Software Engineers

❖ Workers behind the Scene

- Those who design and develop the DBMS and System Environment
- They don't use the database for their own purpose

Summary

- ✓ Definitions of Database and Database Applications
- ✓ Basic Definitions
- ✓ Typical DBMS Functionality
- ✓ Example of a Database (UNIVERSITY)
- ✓ Main Characteristics of the Database Approach
- ✓ Advantages of Using the Database Approach
- ✓ When Not to Use Databases
- ✓ Database Users

End of Chapter One!

Thank You!